

Ch 7: Coordinate Geometry

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Find the distance between the points A(2, 3) and B(4, 1).

(PYQ)

Q2. Find the distance between the points P(-6, 7) and Q(-1, -5).

(PYQ 2019)

Q3. Find the distance between the points A(0, 0) and B(36, 15).

(PYQ)

Q4. Find the distance of the point (-3, 4) from the x-axis. (PYQ

2012)

Q5. Find the distance of the point (5, -12) from the origin.

(PYQ)

Q6. Find the distance of the point (a cos θ , a sin θ) from the

origin. (PYQ)

Q7. Find the value of x such that PQ = QR where P = (6, -1), Q

= (1, 3), R = (x, 8). (PYQ 2020)

Q8. Find the value of a such that the distance between A(a, -3)

and B(-4, 7) is 10. (PYQ)

Q9. Find the value of y if the distance between $A(0, 0)$ and $B(0, y)$ is 6. (PYQ)

Q10. Check whether the points $(1, 5)$, $(2, 3)$, and $(-2, -11)$ are collinear. (PYQ)

Q11. Check whether the points $(3, 2)$, $(-2, -3)$, and $(2, 3)$ form a triangle. (PYQ)

Q12. Find the point on the x -axis which is equidistant from $(2, -5)$ and $(-2, 9)$. (PYQ 2019)

Q13. Find the point on the y -axis equidistant from $(6, 5)$ and $(-4, 3)$. (PYQ)

Q14. If $A(4, 3)$, $B(-1, y)$, and $C(3, 4)$ are vertices of a right triangle ABC right-angled at A , find y . (PYQ 2015)

Q15. Find the ratio in which the point $(-3, p)$ divides the line segment joining $A(-5, -4)$ and $B(-2, 3)$. Hence find p . (PYQ 2020)

Q16. Find the ratio in which the y -axis divides the line segment joining $A(5, -6)$ and $B(-1, -4)$. Also find the coordinates of the point of division. (PYQ 2019)

Q17. Find the ratio in which the x -axis divides the line segment joining $A(2, -3)$ and $B(5, 6)$. Also find the coordinates of the point. (PYQ 2024)

Q18. Find the ratio in which the line segment joining the points $(5, 3)$ and $(-1, 6)$ is divided by the y -axis. (PYQ 2024)

Q19. Find the coordinates of the midpoint of the segment joining $A(-2, 3)$ and $B(4, -5)$. (PYQ)

Q20. Find the coordinates of the point which divides the line joining $(1, -3)$ and $(-3, 9)$ in ratio $1:3$. (PYQ)

Q21. Find the coordinates of the point which divides the line segment joining $A(4, -3)$ and $B(8, 5)$ in ratio $3:1$ internally. (PYQ 2019)

Q22. Find the coordinates of the centroid of triangle with vertices $A(-2, 3)$, $B(6, 7)$, and $C(4, 2)$. (PYQ)

Q23. The midpoint of segment AB is $P(0, 4)$. If $A = (-2, b)$ and $B = (2, c)$, find $b + c$. (PYQ)

Q24. Find the coordinates of the points of trisection of the segment joining $A(2, -2)$ and $B(-7, 4)$. (PYQ 2020)

Q25. If the midpoints of the sides of a triangle are $(1, 2)$, $(0, -1)$, and $(2, -1)$, find the vertices of the triangle. (PYQ)

Q26. If $(1, 2)$, $(4, y)$, $(x, 6)$, and $(3, 5)$ are the vertices of a parallelogram, find x and y . (PYQ 2018)

Q27. Find the value of k if the points $A(2, 3)$, $B(4, k)$, and $C(6, -3)$ are collinear. (PYQ 2020)

Q28. If $A(-2, 4)$, $B(0, 0)$, and $C(4, 2)$ are the vertices of triangle ABC , find the length of the median through A . (PYQ)

Q29. Show that the points $A(1, 7)$, $B(4, 2)$, $C(-1, -1)$, and $D(-4, 4)$ are vertices of a square. (PYQ 2019)

Q30. The centre of a circle is $(2\alpha - 1, 7)$ and it passes through the point $(-3, -1)$. If the diameter is 20 units, find α . (PYQ)

SECTION B - 3 MARKS

Q1. Prove that the points $(3, 0)$, $(6, 4)$, and $(-1, 3)$ are the vertices of a right-angled isosceles triangle. (PYQ 2019)

Q2. Prove that the points $(2, -2)$, $(-2, 1)$, and $(5, 2)$ are the vertices of a right-angled triangle. Find its area. (PYQ)

Q3. Show that the points $A(5, 6)$, $B(1, 5)$, $C(2, 1)$, and $D(6, 2)$ are the vertices of a square. (PYQ 2020)

Q4. Show that the points $A(1, -2)$, $B(3, 6)$, $C(5, 10)$, and $D(3, 2)$ are the vertices of a parallelogram. (PYQ)

Q5. Show that the points $A(0, -2)$, $B(3, 1)$, $C(0, 4)$, and $D(-3, 1)$ are the vertices of a square. (PYQ 2018)

Q6. Show that $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(7, 3)$ are the vertices of a rhombus. (PYQ 2019)

Q7. If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(k, 3)$ are the vertices of a parallelogram, find k . (PYQ 2019)

Q8. Show that the quadrilateral with vertices $A(-1, -2)$, $B(1, 0)$, $C(-1, 2)$, and $D(-3, 0)$ is a square. (PYQ)

Q9. The x-coordinate of a point P is twice its y-coordinate. If P is equidistant from Q(2, -5) and R(-3, 6), find the coordinates of P. (PYQ 2020)

Q10. Find the value of k if the point P(2, 4) is equidistant from A(5, k) and B(k, 7). (PYQ 2019)

Q11. If the distances of P(x, y) from A(5, 1) and B(-1, 5) are equal, prove that $3x = 2y$. (PYQ)

Q12. Find the ratio in which the line segment joining A(1, -5) and B(-4, 5) is divided by the x-axis. Also find the coordinates of the point of division. (PYQ 2020)

Q13. Find the ratio in which the line $2x + y - 4 = 0$ divides the line segment joining A(2, -2) and B(3, 7). (PYQ 2019)

Q14. Find the ratio in which the point (-3, k) divides the segment joining (-5, -4) and (-2, 3). Hence find k. (PYQ 2020)

Q15. In what ratio does the point $(\frac{2}{3}, y)$ divide the segment joining (-2, 2) and (3, 7)? Also find y. (PYQ)

Q16. Find the coordinates of the points of trisection of the line segment joining A(4, -1) and B(-2, -3). (PYQ 2018)

Q17. Find the coordinates of the points which divide the segment joining A(-2, 2) and B(2, 8) into four equal parts. (PYQ)

Q18. If A(-2, -1), B(a, 0), C(4, b), and D(1, 2) are the vertices of a parallelogram, find a and b. (PYQ 2019)

Q19. The midpoints of the sides AB, BC, and CA of triangle ABC are D(3, 4), E(8, 9), and F(6, 7). Find the vertices A, B, and C. *(PYQ)*

Q20. Find the value of k for which the points (7, -2), (5, 1), and (3, k) are collinear. *(PYQ 2020)*

Q21. If A(-5, 7), B(-4, -5), C(-1, -6), and D(4, 5) are the vertices of a quadrilateral, find its area. *(PYQ)*

Q22. Find the centre of the circle passing through (6, -6), (3, -7), and (3, 3). *(PYQ 2019)*

Q23. Show that the points A(-1, 0), B(3, 1), C(2, 2), and D(-2, 1) are vertices of a parallelogram. Find whether it is a rectangle. *(PYQ)*

Q24. Find the third vertex of a triangle whose two vertices are (-4, 1) and (0, -3) and centroid is at (2, -1). *(PYQ 2020)*

Q25. If two vertices of an equilateral triangle are (0, 0) and (3, $\sqrt{3}$), find the third vertex. *(PYQ)*

Q26. Let P and Q be the points of trisection of the segment joining A(2, -2) and B(-7, 4) such that P is nearer to A. Find coordinates of P and Q. *(PYQ 2019)*

Q27. The line joining the points A(4, -5) and B(4, 5) is divided by the point P such that $AP/AB = 2/5$. Find the coordinates of P. *(PYQ)*

Q28. Find the ratio in which the segment joining (-2, -3) and (5, 6) is divided by: (i) x-axis (ii) y-axis. *(PYQ)*

Q29. If $A(4, 2)$, $B(7, 6)$, and $C(1, 4)$ are the vertices of triangle ABC and AD is the median, find the length of AD . *(PYQ 2019)*

Q30. Two friends start from opposite ends of a road 10 km long. If they meet at a point which divides the road in ratio 3:2 from one end, find the meeting point if the road starts at $(0, 0)$ and ends at $(10, 10)$. *(PYQ real-life type)*

Q31. Show that $A(3, 2)$, $B(0, 5)$, $C(-3, 2)$, and $D(0, -1)$ are the vertices of a square. Find the area using the distance formula. *(PYQ)*

Q32. If $A(1, 2)$, $B(4, 3)$, and $C(6, 6)$ are vertices of a triangle, find whether the median from A bisects the opposite side BC . *(PYQ)*

Q33. Find the point on the x -axis which is equidistant from $A(3, 4)$ and $B(-5, 6)$. *(PYQ)*

Q34. If the centroid of a triangle formed by (a, b) , (b, c) , and (c, a) is the origin, find $a + b + c$. *(PYQ)*

Q35. Find the ratio in which the segment joining $(4, 8)$ and $(5, 5)$ is divided by the point $(7/2, 6)$. Verify using the section formula. *(PYQ)*

SECTION C - 5 MARKS

Q1. Find the ratio in which the y-axis divides the segment joining $(5, -6)$ and $(-1, -4)$. Also find the point of intersection. Then verify by showing both distances using the distance formula. (PYQ 2019)

Q2. In what ratio does the x-axis divide the segment joining $(2, -3)$ and $(5, 6)$? Find the coordinates of the point. Hence find the distance of this point from the origin. (PYQ 2024)

Q3. Find the coordinates of the points of trisection of the line segment joining $A(2, -2)$ and $B(-7, 4)$. Hence verify that these points lie on AB. (PYQ 2020)

Q4. Show that $A(1, 7)$, $B(4, 2)$, $C(-1, -1)$, and $D(-4, 4)$ are vertices of a square. Find the diagonal length and verify that the diagonals are equal. (PYQ 2019)

Q5. Show that $A(5, 6)$, $B(1, 5)$, $C(2, 1)$, and $D(6, 2)$ are the vertices of a square. Find the length of diagonals and verify that they are equal. (PYQ 2020)

Q6. Prove that the points $(2, 3)$, $(-4, -6)$, and $(1, 3/2)$ are collinear. Also find the ratio in which the middle point divides the segment. (PYQ)

Q7. Show that $A(0, -2)$, $B(3, 1)$, $C(0, 4)$, and $D(-3, 1)$ form a square. Find: (i) all four sides (ii) both diagonals and verify $AC = BD$. (PYQ 2018)

Q8. The midpoints of the sides of a triangle are $(1, 5/2)$, $(0, 1/2)$, and $(1/2, 0)$. Find the vertices of the triangle. *(PYQ)*

Q9. If the points $A(k + 1, 2k)$, $B(3k, 2k + 3)$, and $C(5k - 1, 5k)$ are collinear, find k . *(PYQ 2019)*

Q10. If $A(-2, 1)$, $B(a, 0)$, $C(4, b)$, and $D(1, 2)$ are vertices of a parallelogram, find a and b . Hence find the lengths of all four sides and verify that opposite sides are equal. *(PYQ 2019)*

Q11. The vertices of a triangle ABC are $A(4, 6)$, $B(1, 5)$, and $C(7, 2)$. A line is drawn to intersect sides AB and AC at D and E respectively such that $AD/AB = AE/AC = 1/4$. Find the lengths of DE , BC and show that $DE = (1/4)BC$. *(PYQ NCERT)*

Q12. Find the point P on the x -axis equidistant from $A(7, 6)$ and $B(-3, 4)$. Hence find the area of triangle PAB . *(PYQ)*

Q13. If P divides the line segment joining $A(-1, 3)$ and $B(9, 8)$ in ratio $k:1$, find P in terms of k . If P lies on the line $x - y + 2 = 0$, find k . *(PYQ 2020)*

Q14. The line segment joining $A(-2, 9)$ and $B(6, 3)$ is a diameter of a circle with centre C . Find C . Also find the radius and the equation of the circle passing through A , B . *(PYQ)*

Q15. If three consecutive vertices of a parallelogram $ABCD$ are $A(1, 2)$, $B(3, 6)$, and $C(5, 10)$, find the fourth vertex D . Verify using the diagonal midpoint property. *(PYQ)*

Q16. Find the coordinates of a point A where AB is diameter of circle whose centre is $(2, -3)$ and B is $(1, 4)$. Also verify that A lies on the circle by checking the distance from centre. (PYQ)

Q17. The two opposite vertices of a square are $(-1, 2)$ and $(3, 2)$. Find the coordinates of the other two vertices. (PYQ 2020)

Q18. An equilateral triangle has two vertices at $(0, 0)$ and $(3, 0)$. Find the third vertex. Then find all three sides and verify the triangle is equilateral. (PYQ)

Q19. The base QR of an equilateral triangle PQR lies on the x-axis. $Q = (-4, 0)$ and the origin is the midpoint of QR. Find the coordinates of P and R. (PYQ 2019)

Q20. If the distance of $P(x, y)$ from $A(3, 6)$ and $B(-3, 4)$ are equal, show that $3x + y = 5$. Hence find the point on the line $3x + y = 5$ that is equidistant from A and B and also lies on the y-axis. (PYQ HOTS)

Q21. Case Study: The map of a triangular park has vertices at $A(1, 7)$, $B(-5, 5)$, and $C(-1, -1)$. A lamp post is to be installed at the centroid of the park. Find: (i) the centroid (ii) the distance of centroid from each vertex (iii) which vertex is nearest to the centroid. (PYQ 2023 case-based type)

Q22. Case Study: A school building has corners at $P(3, 4)$, $Q(6, 7)$, $R(9, 4)$, and $S(6, 1)$. Find: (i) whether PQRS is a square or rhombus (ii) the length of diagonals (iii) ratio in which diagonal PR is divided by diagonal QS. (PYQ 2024 case-based type)